

Spectral Token Mixing

Frequency-Domain Speed Optimization for Long-Context LLMs



Attention is the speed limit of large language models — its $O(n^2)$ mixing step owns up to ~95% of long-context compute. This paper shows how to lift that limit: **transform to frequency space, mix in $O(n)$, transform back**. We develop why sub-quadratic attention is mathematically possible, a staged path from a 30-line reference operator to fused tensor-core kernels, the validation gates any claim must pass, and what an up-to-**18x inference speedup** is worth in the real world.

The one-paragraph version. Self-attention is quadratic because it explicitly computes all n^2 pairwise token interactions. That cost is a *representation choice*, not a law of nature. Any mixing that depends on relative position is a convolution, and by the convolution theorem a convolution becomes **pointwise multiplication** after a Fourier transform: n^2 multiply-adds collapse to n , plus two FFTs at $n \log n$. The frequency↔time conversion is not overhead — it *is* the algorithm. What cannot be made exact (content-based lookup) is approximated, bounded, and measured — never assumed.

01 Where $O(n^2)$ actually comes from

One attention layer computes, per head: $\text{Attention}(Q, K, V) = \text{softmax}(QK^T/\sqrt{d}) \cdot V$. Everything expensive lives in one object — the **$n \times n$ mixing matrix**: row i says how much token i listens to every other token. Computing QK^T costs $2n^2d$ FLOPs, applying it to V costs another $2n^2d$, and at long context the n^2 memory traffic dominates latency while the KV cache dominates VRAM.

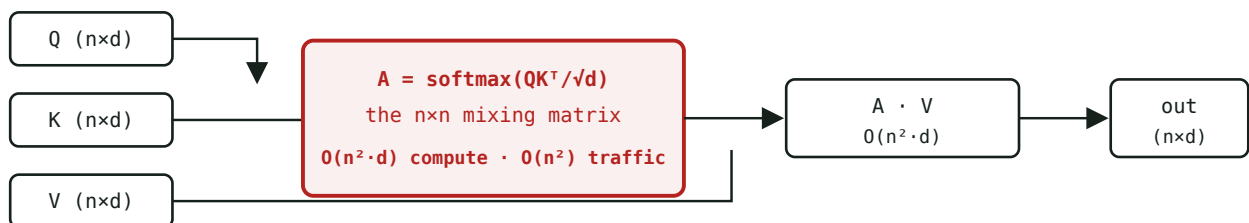


Figure 1 — the bottleneck is one object. Input and output are both $(n \times d)$; the quadratic cost exists only in the middle. That is exactly what makes a drop-in replacement mathematically possible.

The crucial observation: **the model never needs the matrix — it needs the matrix's action on $\mathbf{V}$** . The n^2 cost is the price of the most general possible representation (every entry independent). Any structure in the mixing is an opportunity to apply it more cheaply.

02 Why sub-quadratic is possible at all

2.1 The change-of-basis principle

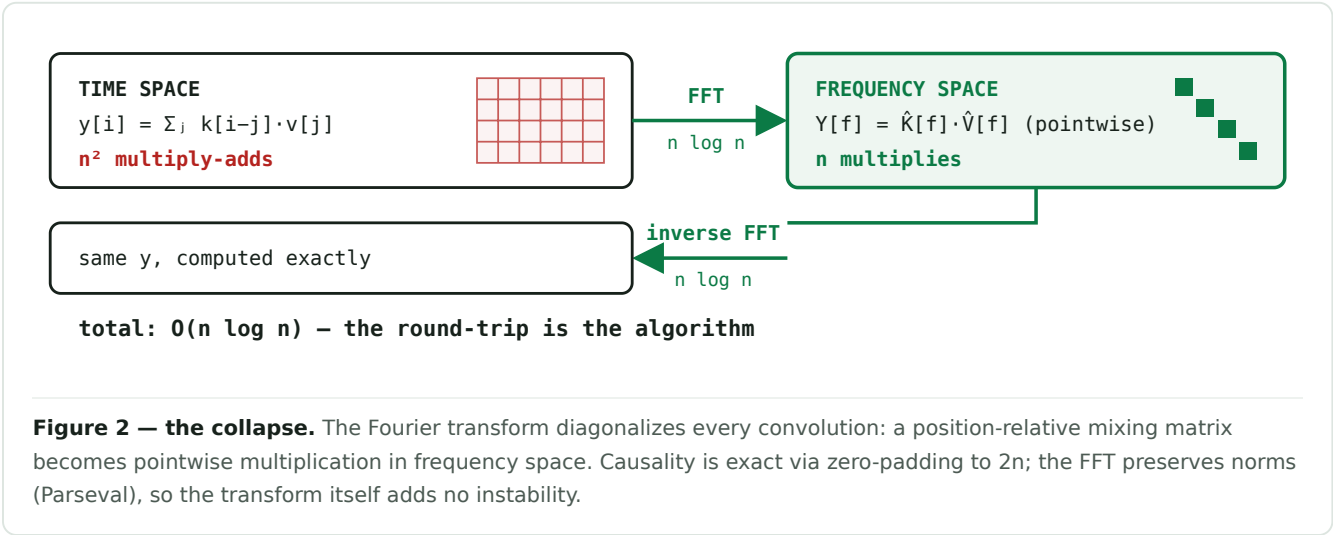
A dense matrix that looks unstructured in one basis can be trivial in another. Every known escape below $O(n^2)$ is an instance of one principle:

Find a basis where the mixing operator is (approximately) structured; transform in, apply the cheap structured form, transform out.

STRUCTURE OF A	BASIS WHERE IT IS CHEAP	APPLY COST	LINEAGE
Low-rank: $A \approx U \cdot W^T$	any rank- r sketch	$O(n \cdot d \cdot r)$	linear attention, Performer
Convolutional / Toeplitz: $A[i,j] = f(i-j)$	Fourier (frequency) basis	$O(n \log n)$	FNet, H3, Hyena, SSMs — this paper
Banded / local + few globals	time basis (sparsity)	$O(n \cdot w)$	sliding-window, Longformer-class

Attention matrices in trained LLMs are empirically *dominated* by exactly these components: strong local bands (syntax), smooth position-relative decay (recency), plus a thin data-dependent residual (retrieval heads). That decomposition is what the hybrid design in §4.2 exploits.

2.2 The convolution theorem — the engine of the frequency route



2.3 The honest boundary — what theory forbids

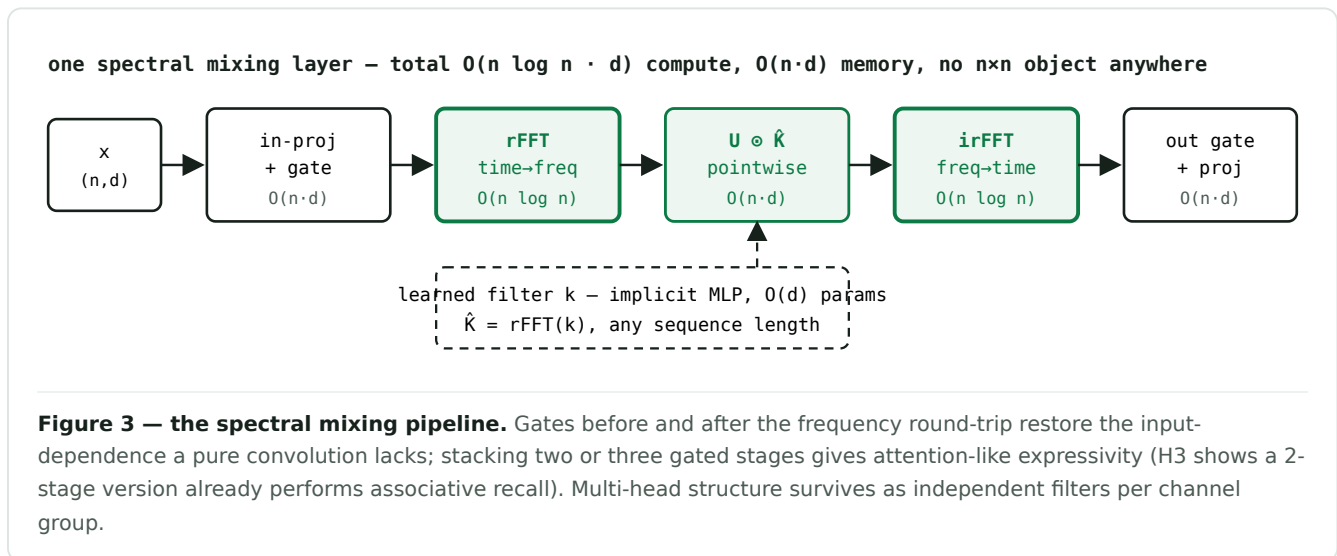
- **Exact attention is quadratic in the worst case.** Under the Strong Exponential Time Hypothesis, no algorithm computes softmax attention exactly in sub-quadratic time for arbitrary inputs (Keles, Wijewardena & Hegde, 2022).

- **Approximation is possible precisely when entries are bounded.** Alman & Song (2023) show near-linear-time attention *approximation* exists when query/key magnitudes are bounded — and cannot exist when they are not.

So "under $O(n^2)$ " is never free equivalence. It is **structure + bounded error**, and the error must be measured against the exact operator on the same inputs — never assumed.

03 The idea in detail: spectral token mixing

Replace (or complement) the data-dependent mixing $\text{softmax}(QK^T) \cdot V$ with a **long convolution applied in frequency space**, made data-dependent by cheap elementwise gates (the H3/Hyena recipe):



What is kept, what is approximated

Kept exactly: position-relative mixing of any shape and range (the filter is length n — a global receptive field from layer one), causality, norm stability, streaming-friendly linearity.

Approximated: content-based addressing — "attend to the token that says X, wherever it is." A convolution cannot do this alone; the gates recover much of it; the hybrid (§4.2) and calibration close the rest. This is the honest reason FNet (pure Fourier mixing, no gates) reaches only ~92–97% of BERT quality, while gated long-convolution designs (Hyena) match Transformer perplexity at scale.

04 Implementation — the process in detail

4.1 Stage 1 — the reference operator (PyTorch, weeks)

A complete, runnable spectral mixer is ~30 lines:

```

class SpectralMixer(nn.Module):
    # x: (batch, n, d)
    def __init__(self, d, filter_mlp_width=64):
        super().__init__()
        self.in_proj = nn.Linear(d, 2 * d) # value + gate
        self.out_proj = nn.Linear(d, d)
        self.filter = ImplicitFilter(d, filter_mlp_width) # pos -> k[pos]

    def forward(self, x):
        n = x.shape[1]
        v, g = self.in_proj(x).chunk(2, dim=-1)
        u = v * torch.sigmoid(g) # gated input
        k = self.filter(n) # (n, d) learned filter
        U = torch.fft.rfft(u, n=2 * n, dim=1) # zero-pad -> causal conv
        Kf = torch.fft.rfft(k, n=2 * n, dim=0)
        y = torch.fft.irfft(U * Kf, n=2 * n, dim=1)[:n]
        return self.out_proj(y * torch.sigmoid(g)) # output gate

```

Fix the evaluation shapes once (e.g. d_{model} 4096, 32 heads, sequence lengths 8k/32k/128k, seeded random Q/K/V from a reference model), then measure four things against the exact attention output **on identical inputs**: MSE, wall-clock latency, peak VRAM, and the scaling curve. Raw spectral mixing will *not* match exact attention zero-shot — record that honestly; it is the baseline the next stage improves on.

4.2 Stage 2 — hybrid + calibration (1-2 months)

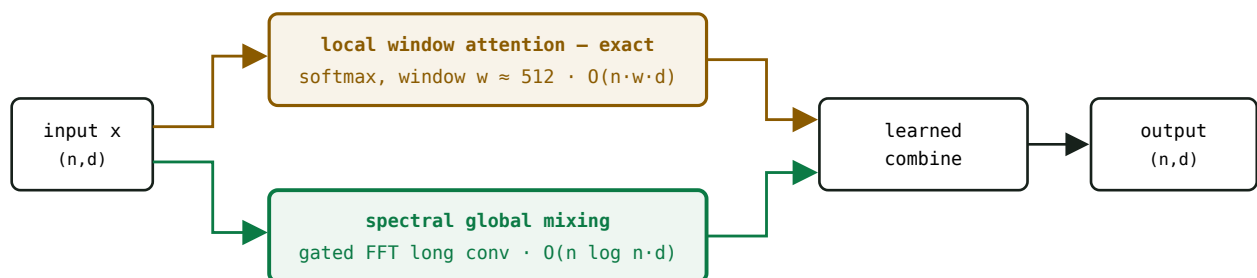


Figure 4 — the hybrid operator. Total cost $O(n \cdot (w + \log n) \cdot d)$ — sub-quadratic. The exact local band handles syntax and recency, where most of the mixing mass and most of the error lives; the spectral path carries the global, long-range component.

Two moves close most of the accuracy gap:

- **Hybrid**: exact sliding-window attention for the local band; spectral for the global remainder. Window cost $O(n \cdot w)$ with $w \approx 512$ stays sub-quadratic.
- **Calibration / distillation**: fit the filters, gates, and combine weights to *reproduce the exact layer's outputs* on sampled inputs — the teacher is `softmax(QKT)V` itself, so no labeled data is needed. LoLCATs (2024) demonstrates that linearizing a full Llama-scale model this way recovers quality while training <0.2% of parameters (LoRA-sized patches).

4.3 Stage 3 — kernels and the decode path (2-6 months)

- **Fused FFT-conv kernel**: one CUDA/Triton kernel doing pad → rFFT → pointwise → irFFT → gate without touching HBM in between. The state of the art is a **Monarch decomposition of the FFT** (FlashFFTConv, 2023): the FFT rewritten as chained small dense matmuls that run on **tensor cores** — the same silicon that makes attention fast — with up to ~8× over stock FFT convolution and long-context wall-clock wins over FlashAttention-2.

- **Precision plan:** fp16/bf16 storage; butterfly/Monarch stages accumulate in fp32. FFT roundoff grows only $O(\sqrt{\log n})$ — benign — but the pointwise stage on a $2n$ -length spectrum needs fp32 accumulation to hold tight MSE at 128k and beyond.
- **Decode path (the honest hard part):** FFT convolution is a *prefill* operator; naively re-running it per generated token would cost $O(n \log n)$ per token. The production answers, in order of maturity: (a) chunked/blocked FFT decode with cached spectra; (b) distilling the fitted filters into a **state-space (recurrent) form** with $O(1)$ per-token state — exact for the filter class the implicit parameterization produces. This is a required deliverable, not an afterthought.

4.4 Validation protocol

A claim only counts when it survives all four gates, measured on pinned hardware against the exact operator on identical inputs:

GATE	TEST	PASS CONDITION
Accuracy	MSE vs exact attention output	below the target budget (e.g. <0.01 at 32k, tightening with scale)
Cost dominance	latency + peak VRAM vs exact	both strictly lower
Scaling proof	run at n and $2n$	latency ratio $\approx 2.1\text{--}2.3\times$, not $4\times$
Compounding	full-model spot check: a fact planted deep in a long sequence must be retrieved through all layers	retrieval holds; per-layer error does not amplify

The scaling proof certifies "sub-quadratic" as a *measured* property rather than a claim; the compounding check protects against a per-layer error that looks small but destroys the model over 32–80 layers.

05 The accounting — verified numbers

FLOP model for a 70B-class transformer (d_{model} 8192, 80 layers; linear parts $\approx 29 \cdot d^2$ FLOPs/token/layer covering QKVO projections + SwiGLU FFN; quadratic mixing = $4n^2 \cdot d$ per layer; spectral mixing costed generously at $10 \cdot n \cdot \log_2 n \cdot d$ per layer):



Figure 5 — where prefill compute goes, and what the spectral route does. Below ~16k the FFN dominates and the spectral route is nearly irrelevant — this is why short-context benchmarks under-sell it. The quadratic term crosses 50% of compute around $n \approx 60k$ and owns the budget from there.

CONTEXT N	MIXING SHARE OF PREFILL (EXACT)	SPECTRAL CUT OF THE MIXING TERM	TOTAL PREFILL SPEEDUP
8k	12%	252×	1.1×
128k	69%	3,084×	3.2×
1M	95%	20,972×	18.6×

Memory: no $n \times n$ object at any point; activation memory $O(n \cdot d)$. At decode, the distilled recurrent form replaces the KV cache — **~43 GB at 128k context** (fp16, 8x GQA, on the model above) — with a **constant-size state**.

06 Real-world impact

In GPU economics. Prefill cost at 128k context drops ~3x; at 1M, ~18x. For an inference provider whose long-context traffic is prefill-dominated (document analysis, code-repository ingestion, agent memory replay), that is the difference between serving a 1M-token request in minutes versus tens of seconds on the same silicon — or serving it at all: at today's frontier pricing a single 1M-token prompt costs dollars; an 18x compute cut moves it toward dimes. The KV-cache elimination compounds this at decode: constant state means long conversations stop evicting batch slots, which raises tokens/sec/GPU — the metric that *is* the business.

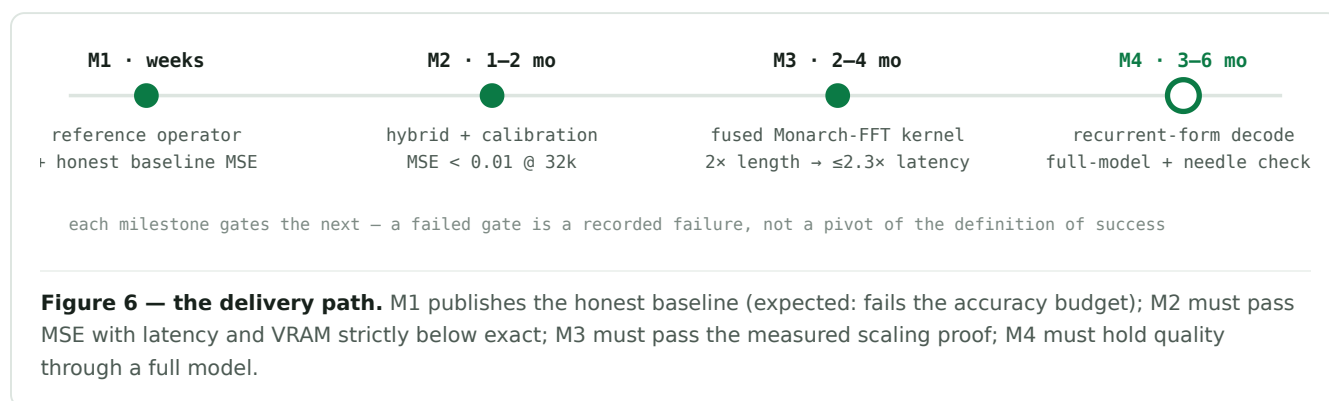
In what becomes possible. Capabilities that today are simulated with retrieval and chunking become native: whole-codebase reasoning, full-day agent sessions with unbroken memory, entire legal/medical corpora in one window, hour-scale audio and video streams. Every one of these is currently gated not by model quality but by the quadratic bill.

On the edge. Constant decode state is the enabler for *local* long-memory agents on consumer and embedded GPUs, where a 43 GB KV cache simply does not fit. A sub-quadratic mixer is the difference between edge devices running short-context toys and running genuine assistants.

In energy. Prefill FLOPs are watts. An 18× cut at million-token scale is not an optimization; it is the difference between long-context AI being an energy story regulators notice and one they don't.

Impact honesty box. The numbers above are ceilings conditional on the gates in §4.4: they assume the hybrid, calibrated operator passes its MSE budget at target context, that the recurrent distillation holds decode quality (needle-in-a-haystack retrieval is the known weak spot of convolutional mixers — it must be in the test suite), and that the fused kernel realizes its FLOP advantage in wall-clock (memory-bound regimes can eat asymptotic wins; the scaling proof exists to catch this). A failed gate is a recorded failure, not a footnote.

07 Milestones



08 References

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8. Alman, Song — *Fast Attention Requires Bounded Entries* (2023)
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10. Zhang, et al. — *LoLCATs: On Low-Rank Linearizing of Large Language Models* (2024)

Working draft. The figures in §5–6 are analytic estimates from the stated FLOP model; they become claims only when reproduced as hardware measurements under the validation protocol of §4.4. This document describes a research and engineering program; it is not an offer of securities and nothing herein constitutes financial advice.